



Thermology

热学

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容与

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Typeset with L^AT_EX

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1 导论

温标

$$T(p) = 273.16 \text{ K} \cdot \frac{p}{p_{\text{tr}}}$$

其中 p_{tr} 为气体在 273.16 K 的压强.

理想气体状态方程

$$pV = \nu RT \quad p = nkT$$

其中普适气体常量: $R = 8.314 \text{ J}/(\text{mol} \cdot \text{K})$, Boltzmann 常量: $k = \frac{R}{N_A} = 1.38 \times 10^{-23} \text{ J/K}$.

理想气体压强公式

$$p = \frac{2}{3} n \bar{\varepsilon}_t$$

分子热运动平均平动动能

$$\bar{\varepsilon}_t = \frac{1}{2} m \overline{v^2} = \frac{3}{2} kT$$

理想气体分子间平均距离

$$d = \sqrt[3]{\frac{1}{n}}$$

Van der Waals 方程

$$\left(p + \frac{a}{V_m^2} \right) (V_m - b) = RT$$

1. 分子固有体积修正: $p \rightarrow \infty, V_m \rightarrow b$;

2. 分子间吸引力修正: $\Delta p_i = \frac{2\Delta k n \bar{v}}{6} = \frac{n^2 K \bar{v}}{3} = \frac{N_A^2 K \bar{v}}{3V_m^2} = \frac{a}{V_m^2}, p_{\text{内}} = p_{\text{壁}} + \frac{a}{V_m^2}$.

非 1 mol 气体的 Van der Waals 方程

$$\left(p + \nu^2 \frac{a}{V^2} \right) (V - \nu b) = \nu RT$$

2 平衡态分子动理论

Maxwell 速率分布

$$f(v) = 4\pi \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}}$$

典型速率值

- 最概然速率

$$v_p = \sqrt{\frac{2kT}{m}}$$

- 平均速率

$$\bar{v} = \sqrt{\frac{8kT}{\pi m}}$$

- 方均根速率

$$v_{\text{rms}} = \sqrt{\bar{v}^2} = \sqrt{\frac{3kT}{m}}$$

气体分子碰壁数

$$\Gamma = \frac{1}{4} n \bar{v}$$

能量均分原理

每个自由度的平均动能为 $\frac{1}{2}kT$.

理想气体内能

$$U = \nu \cdot \frac{i}{2} RT$$

$$dU = C_V dT = \nu C_{V,m} dT$$

3 非平衡态理论

Newton 黏性定律

黏性力

$$F = -\eta \frac{du}{dz} S$$

动量流密度

$$J_P = -\eta \frac{du}{dz}$$

热传导 Fourier 定律

热流

$$\frac{dQ}{dt} = -\kappa \frac{dT}{dz} S$$

热流密度

$$J_T = -\kappa \frac{dT}{dz}$$

碰撞截面

$$\sigma = \pi d^2 = \frac{1}{4} \pi (d_1^2 + d_2^2)$$

平均碰撞频率

$$\bar{Z} = n \bar{v}_{12} \sigma = \sqrt{2} n \bar{v} \sigma$$

平均自由程

$$\bar{\lambda} = \frac{1}{\sqrt{2} n \sigma} = \frac{kT}{\sqrt{2} \sigma p}$$

4 热力学第一定律

热力学第一定律

$$Q = \Delta U + A$$

$$\mathrm{d}Q = \mathrm{d}U + p \mathrm{d}V$$

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$$H = U + pV = U + \nu RT$$

等温过程的热量、内能、做功

$$Q = \nu RT \ln \frac{V_2}{V_1} = \nu RT \ln \frac{p_1}{p_2}$$

$$\Delta U = 0$$

$$A = \nu RT \ln \frac{V_2}{V_1} = \nu RT \ln \frac{p_1}{p_2}$$

等体过程的热量、内能、做功

$$Q = \nu C_{V,m} \Delta T$$

$$\Delta U = \nu C_{V,m} \Delta T$$

$$A = 0$$

等压过程的热量、内能、做功

$$Q = \nu(C_{V,m} + R)\Delta T = \nu C_{p,m} \Delta T$$

$$\Delta U = \nu C_{V,m} \Delta T$$

$$A = p\Delta V = \nu R\Delta T$$

绝热方程

$$pV^\gamma = C$$

其中, 比热容比:

$$\gamma = \frac{C_{p,m}}{C_{V,m}} = \frac{i+2}{i}$$

多方过程方程

$$pV^n = C_1 \quad TV^{n-1} = C_2 \quad T^n p^{1-n} = C_3$$

$$\begin{cases} \text{等温: } pV = C & \Rightarrow n = 1 \\ \text{等体: } V = C & \Rightarrow n \rightarrow \infty \\ \text{等压: } p = C & \Rightarrow n = 0 \\ \text{绝热: } pV^\gamma = C & \Rightarrow n = \gamma \end{cases}$$

绝热过程的热量、内能、做功

$$Q = 0$$

$$\Delta U = \nu C_{V,m} \Delta T = \frac{p_2 V_2 - p_1 V_1}{\gamma - 1} = \frac{\nu R (T_2 - T_1)}{\gamma - 1}$$

$$A = \frac{p_2 V_2 - p_1 V_1}{1 - \gamma} = \frac{\nu R (T_2 - T_1)}{1 - \gamma}$$

多方过程的热量、内能、做功

$$Q = \nu \left(C_{V,m} - \frac{R}{n-1} \right) \Delta T = \nu \frac{\gamma - n}{1 - n} C_{V,m} \Delta T$$

$$\Delta U = \nu C_{V,m} \Delta T$$

$$A = \frac{p_2 V_2 - p_1 V_1}{1 - n} = \frac{\nu R (T_2 - T_1)}{1 - n}$$

可逆过程的摩尔热容

- 等温

$$C = \infty$$

- 等容

$$C_{V,m} = \frac{i}{2} R = \frac{R}{\gamma - 1}$$

- 等压

$$C_{p,m} = \frac{i+2}{2} R = \frac{\gamma R}{\gamma - 1}$$

- 绝热

$$C = 0$$

- 多方

$$C_{n,m} = \frac{\gamma - n}{1 - n} C_{V,m}$$

热机效率

$$\eta = \frac{A_{\text{净}}}{Q_{\text{吸}}} = 1 - \frac{Q_{\text{放}}}{Q_{\text{吸}}}$$

Carnot 循环

$$\eta_{\text{C}} = 1 - \frac{T_{\text{低温}}}{T_{\text{高温}}}$$

制冷效率

$$w = \frac{Q_{\text{吸}}}{A_{\text{净}}} = \frac{Q_{\text{吸}}}{Q_{\text{放}} - Q_{\text{吸}}}$$

Carnot 逆循环

$$w_{\text{C}} = \frac{T_{\text{低温}}}{T_{\text{高温}} - T_{\text{低温}}}$$

5 热力学第二定律

Clausius 表述

不可能把热量从低温物体传到高温物体, 而不产生其他影响.

Kelvin 表述

不可能从单一热源吸热使之完全变为功, 而不产生其他影响.

Carnot 定理

$$\eta_{\text{不可逆}} < \eta_{\text{可逆}} = 1 - \frac{T_{\text{低温}}}{T_{\text{高温}}}$$

Clausius 不等式

$$\oint \frac{\mathrm{d}Q}{T} \leq 0$$

Clausius 熵 (热温商)

$$\mathrm{d}S = \frac{\mathrm{d}Q}{T} \quad \Delta S = \int \frac{\mathrm{d}Q}{T}$$

理想气体的熵变计算 $(p_1, V_1, T_1) \rightarrow (p_2, V_2, T_2)$

$$\begin{aligned} S_2 - S_1 &= \nu C_{V,m} \ln \frac{T_2}{T_1} + \nu R \ln \frac{V_2}{V_1} = \nu C_{p,m} \ln \frac{T_2}{T_1} - \nu R \ln \frac{p_2}{p_1} \\ &= \nu C_{V,m} \ln \frac{p_2}{p_1} + \nu C_{p,m} \ln \frac{V_2}{V_1} \end{aligned}$$

可逆过程的熵变

- 等温

$$\Delta S = \nu R \ln \frac{V_2}{V_1}$$

可逆 $\Delta S = \frac{Q}{T}$.

- 等容

$$\Delta S = \nu C_{V,m} \ln \frac{T_2}{T_1}$$

- 等压

$$\Delta S = \nu C_{p,m} \ln \frac{T_2}{T_1}$$

- 绝热 $\Delta S = 0$, 不可逆 $\Delta S \neq 0$.

热力学基本方程

$$dU = T dS - p dV$$

p - V 图 & T - S 图

$$\begin{cases} A = \int p dV & (p-V \text{ 图}) \\ Q = \int T dS & (T-S \text{ 图}) \end{cases}$$

对于循环过程

$$\begin{cases} \oint p dV = A_{\text{净}} \\ \oint T dS = Q_{\text{净}} \end{cases}$$

$$\oint dU = 0 \implies Q_{\text{净}} = A_{\text{净}}$$

即 p - V 图与 T - S 图中封闭曲线所围面积均等于循环过程做的净功, 顺时针为正循环, $A_{\text{净}} > 0$, 净功也等于净吸热量.

熵增原理

绝热独立系统, 经过某一过程, 总体 $\Delta S \geq 0$.

Boltzmann 熵 (Boltzmann 关系)

$$S = k \ln \Omega$$

6 物态与相变

液体表面张力系数

$$F = \sigma l \quad dW = \sigma dS$$

弯曲液面压强差 (Laplace 公式)

$$\Delta p = \frac{2\sigma}{R}$$

毛细现象毛细管内液面高度变化 (Jurin 公式)

$$h = \frac{2\sigma \cos \theta}{\rho g r}$$

Clapeyron 方程

汽化热 L_v , 熔化热 L_m , 升华热 L_s .

- 气液相变

$$\frac{dp}{dT} = \frac{L_v}{T(V_g - V_l)}$$

- 固液相变

$$\frac{dp}{dT} = \frac{L_m}{T(V_l - V_s)}$$

- 固气相变

$$\frac{dp}{dT} = \frac{L_s}{T(V_g - V_s)}$$

饱和蒸汽压与温度的关系式 ($V_g \gg V_l$)

$$\begin{aligned} \frac{dp}{dT} &\approx \frac{L_v}{TV_g} = \frac{pL_v}{\nu RT^2} \\ \Rightarrow \ln \frac{p}{p_0} &= -\frac{L_v}{\nu R} \left(\frac{1}{T} - \frac{1}{T_0} \right) \end{aligned}$$

Maxwell 关系式

$$\left(\frac{\partial T}{\partial V}\right)_S = \left(\frac{\partial p}{\partial S}\right)_V$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$